# Solutions

1.

a)

i) How are transients in power networks characterised? [2]

*Normally transients are associated with current and voltages- they are caused mainly by lightning and switchings and faults. Usually current is expressed in kA and microseconds, voltage is expressed in kV and ms.*

*A large majority answered this*

ii) Why in extra high voltage (EHV) transmission system, do voltage transients receive the most important consideration? [3]

*At extra higher voltages insulation of various components such as breaker, switch , transformer bus bar, generator are subjected to stress – high voltage stress lead to failure of the component – so such high transient over voltages are main consideration for designing insulation level of various components facing the over-voltage.*

*Some of them got it the insulation design – most did not*

b)

i) What is sub synchronous resonance (SSR) in the context of power system stability? [5]

*The students are expected to state that it is a resonance condition involving two parts of power system at lower than synchronous frequency*

*[2]*

*During network events (switching) high current will flow temporarily through generator stator circuits. Usually these are due to transformer EMF and have power frequency (50/60 Hz). The contribution of this component during transient to total generation voltage is insignificant. Their exclusions from modelling make the analysis simpler to establish better understanding. Various sections of turbine mass dynamically interact with the network particularly when long transmission lines are series compensated. The frequencies of oscillations are in the range of 10 to 48 Hz and that is why they are classified as subsynchronous resonance (SSR). [3]*

*Very few cracked it – it is a difficult concept in stability*

ii) How has the nature of SSR changed with the increasing presence of high-power electronics in the power network? [5]

The student mention the nature of shaft for WTG – how it is different from multi-stage steam turbine .. so what frequency is expected etc..

The electromechanical coupling in renewable is weak to nil – that is how it had to be approached – most of them got it wrong.

[2]

*Then they should indicate control mode interaction which are not torsional in nature but interaction control from different converter from networks; known as converter interaction mode which is also sub-synchronous. Also converter control of HVDC and STATCOM with network series compensation …etc..*

[3]

iii) List three SSR mitigation measures. [3]

*Power system stabiliser, supplementary damping controllers in FACTS, torsional filters, NGH dampers, additional control in converter outer loop control etc..*

*This was fairly straight forward*

c) i) Describe the purpose of having a high gain fast acting excitation system? [3]

*The student is expected to emphasise transient stability of synchronous generator and how suddenly applied large excitation system plays positive role. It is large synchronising torque that is required during the fault to slow down the rotor acceleration dictates the purpose of a fast acting and high gain excitation system, it also helps to increase critical fault clearing time.*

*Many struggle to relate speed of excitation system and transient stability*

ii) Why is a power system stabiliser (PSS) essential when operating a large synchronous generator in power system? [4]

*First students are expected to mention about negative consequence of high gain fast acting AVR [2]*

*The functional requirement of positive damping torque has to be emphasised next. The students are expected to mention the choice of the signals and stabiliser generic structure in terms of phase lag and lead, wash out filter etc. [2]*

[25 marks]

This was answered by large majority

2.

a)

i) The rotor of a synchronous generator is of solid construction whereas the stator is made up of laminated materials. Provide technical reasons to justify such a constructional practice. [4]

*The students are expected to say stator carries power frequency currents – so losses due to hysteresis and eddy currents in the magnetic materials are minimised [2]*

*In the rotor it is DC current – so the relative velocity between rotor and rotor winding is zero – no induced current – similarly impact of stator MMF on the rotor body at synchronous speed is nil – so rotor does not need to be laminated.[2]*

*A large majority answered this*

ii) Besides the excitation winding, the rotor of a synchronous generator usually has additional windings – describe the purpose of such windings. [4]

*The students are expected to demonstrate their understanding of interaction between rotor and stator winding during transient situation i.e non- synchronous speed. [2]*

*One would expect that the frequency of the current is the difference between synchronous frequency and rotor speed. The usefulness of additional winding to provide counter toque to bring the rotor back to initial operating speed. [2]*

*This was answered by more than 70% of the students*

iii) Why is the stator winding in a synchronous generator star connected? [3]

*I am expecting them to talk about enough required current for earth fault protection and grounding requirement*

*It was difficult – very few thought about protection angle..*

iii) Modern synchronous generators are designed to have a large steady state synchronous reactance (between 1.0 to 2.0 pu). However, when a direct terminal short circuit occurs, the initial currents are several times the normal load current. Why is it so? [5]

*The students should demonstrate their understanding that during fault synchronous reactance does not exist [2]*

*They then mention about the electromagnetic flux linkage geometry during fault – sub transients, transients etc and how the varying flux path lead to variable inductance with time .. how that lead to larger currents etc. [3]*

*It was answered by many students*

iv) Why does a synchronous generator offer different overload limits in the lagging and leading power factor mode of operation? [4]

*The flux reinforcement during the leading power factor based synchronous generation saturates the end ring and increases the eddy current … [2]*

*This leads to the losses and heating in the end winding region – thus the permitted output is less than that as lagging power operation. [ 2]*

*Very few brought the end winding heating limit because of flux summation.*

v) Two synchronous generators of identical capacities are delivering similar amount of load to a grid network. One of them is non-salient pole (round rotor) and other one is salient pole (non-round rotor). Excitation is suddenly lost in both cases while the turbine torques continue. Which one will go out of synchronism faster? Justify your answer with reason. [1+4]

*The one with non salient pole [1]*

*The student should be able to explain the saliency power first. [1]*

*The one with salient pole will produce real power because of saliency (difference between d-q axis reactance) – that will lead lesser accelerating torque for the one with salient pole machine. The nonsalient pole machine on the other hand will not able to produce any power during the loss of excitation – so accelerate faster – so the non-salient pole machine is likely to go out of synchronism earlier than the salient pole one [3]*

*More than 80% answered correct*

[25 marks]

3.

a)

i) Why is it difficult to analyse the operating characteristic of a synchronous machine in a stationary 3-phase (a-b-c) reference frame? [3]

*Stator self and mutual inductance in phase domain is function of rotor angle. It is dictated by the loading .. so one needs express inductance which remains fixed – in a-b-c frame it is not possible*

*Many were spot on time and space variation of inductance*

ii) How does a d-q reference frame (Park reference frame) help to overcome such difficulty? [3]

*I am expecting the students to state that the magnetic effect of stator can be characterised by two windings in quadrature and when those two windings are aligned on the polar and*

*interpolar axis – with the same reference as rotor – the inductances are fixed – easy to analyse.*

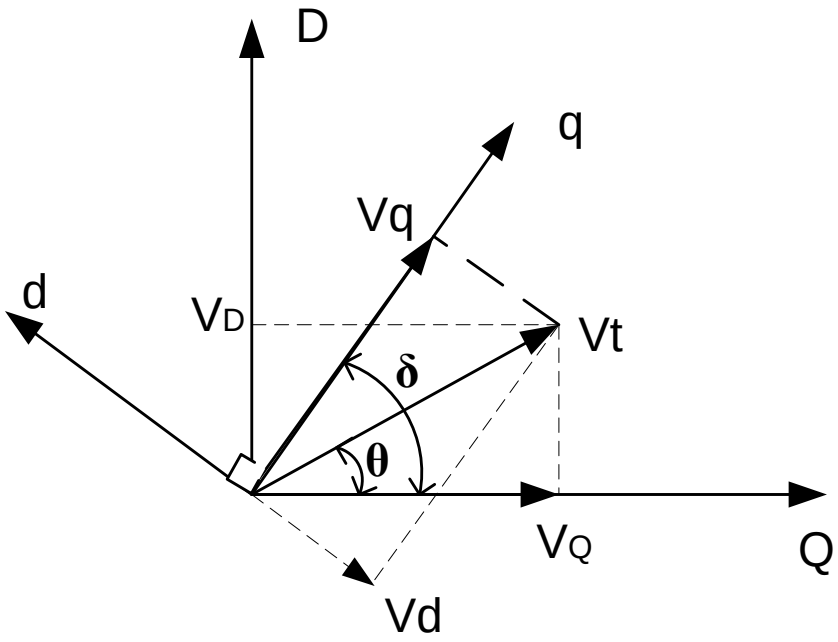
*It was argued to make inductance invariant in space – many students had done this correctly.*

iii) Despite having represented a synchronous machine in d-q reference frame, why another transformation is necessary when the machine is connected to a large power system network? [3]

*Each machine is modelled in its own dq (Park) co-ordinate taking the terminal voltage as reference– the load angle of each machine is different from the other machine – so with respect to the network or Kron reference frame each individual machine has to be represented again requiring machine (Park) to Network (Kron) transformation.*

*Many answered machine network transformation requirement well.*

iii) Derive the relationship between the quantities in two reference frames when the angle difference between the two reference frames is δ-θ. [5]




*[2]*

**Synchronous reference frame with respect to network reference frame**

$$VQ + j\,VD = Vt\,e^{j\theta}$$
$$VQ = Vt\cos\theta$$
$$VD = Vt\sin\theta$$

*Resolving VQ and VD along machine reference d-q will produce*

$$Vq = VQ\,\text{Cos}\,\delta + VD\sin\delta$$
$$Vd = -VQ\sin\delta + VD\cos\delta$$

$$Vq + j\,Vd = Vt\, e^{-j(\delta-\theta)}$$

*Almost all got this right equation and derivation wise , calculation wide some inaccurate results – very gently penalised 1 or 2 marks max*

iv) A synchronous generator supports a voltage of 15.75 kV line to line at the generator terminal bus. This bus is part of a large network which has a reference bus at another location. The power flow solution produces an angle (θ) of 7 degrees for this bus voltage. The generator load angle (δ) is 33 degree. Compute the direct and quadrature axis components of this terminal voltage. [5]

*The students would be able to apply the formula in part (iii) with appropriate choice of angle . in this case Vq+jVd = 15.75*exp(-26 degree)*

*Vq = 15.75*cos26 , Vd =-15.75*sin(-26): Ans Vq=14.15 kV and Vd = -6.9 kV*

*Almost all got this right*

b)

i) Why are Type 3 and Type 4 wind turbine generators preferred over Type 1 and Type 2 in modern wind farms? [3]

*They allow operation at wider variation of wind speed [2]*

*They are more controllable and efficient [1]*

*Almost all got this right*

ii) In a large wind farm, the wind velocity is just 1m/s more than the cut in velocity. There are 10 rows of turbine. It was observed that some turbines in the subsequent rows were not producing power. Explain clearly what could be the reason(s). [3]

*This requires the understanding of wake effect in wind farm – slowing down of wind speed by the turbine up stream – this means wind after heating the first row – will slow down – the speed being very close to cut-in speed – after wake it will be below cut-in value for the next row, so turbine will not start.*

[25 marks]

*Almost all got this right*

4.

a)

i) What is the function of a gearbox in a wind turbine generator? [2]

wind turbine spins at a very low

*Generally, wind turbine rotates at 4-6 RPM depending on the wind speed/radius of the blade. This is way too low to generate electrical power at power frequency such as 50 and or 60 Hz. The shaft to which the generator is connected rotates at high speed. The high shaft speed is produced by the gear box – having lesser teeth on the generator shaft and larger teeth on the turbine side shaft.*

*Almost all got this right*

ii) The gear box is not necessary for some types of wind turbine generators. Explain the reason behind such an option. [2]

*Type 4 employs permanent magnet machine with large number of poles which produces AC waveform between 10-25 Hz. This is then rectified and inverted to grid frequency. This eliminates the need for gear box.*

*Almost all got this right*

iii) How does one model the gear box for stability studies? [2]

*Usually gear box is modelled by a scalar gain. When speed is expressed in per unit, it can even be simplified as unity gain. The shaft on either side can be modelled by spring mass – but that is not part of gear box dynamics*

*Few student answered it right.*

iv) Why does Type 4 wind turbine usually have large number of poles?

[3]

*I expect the student to write along the line like this It is required to have reasonably good frequency of power generation at the stator terminal for optimising magnetics design – in the absence of gear box , large number of poles will produce that frequency at lower wind turbine speed … etc..*

*Nearly half of the student argued it well for full marks*

b)

i) How does stall control differ from pitch control? [3]

*Stall control is realised by increasing the angle of attack – that is more surface area of blade faces the wind. In pitch control , the blade is moved in opposite direction – away from the wind – both result in drop in lift force and hence drop in power production .*

*The students are expected to draw sketch to show the blade position, angle of attack , lift force reduction etc.. ..*

*Almost all got this right*

ii) How is passive stall control implemented? [3]

*In passive-stall control, the angle of attack is designed to work optimally at the rated wind speed. If the wind speed increases above the rated speed, the laminar wind flow above the cord is disrupted, causing the turbulence. This phenomenon is known as stall. As a result, the lift force is reduced with the rotational speed. Passive-stall technique is an effective measure to avoid the turbine from mechanical failure.*

*Almost all got this right*

iii) Why is it necessary to have a delay and dead-band to represent pitch control? [3]

*Too fast pitching is not good for the operating life of blade. Sometimes turbulence/gust in wind may demand frequent pitch control – which is not necessary. There also has to have minimum threshold to make sure pitching is done for appreciable change . etc..*

*Some of them got it right ( less than 30% students)*

iv) Why is generator torque/power control preferred over pitch control at sub-rated wind condition? [3]

*I expect the student to mention important aspect about mechanical control – it is designed to have at higher than rated speed because these controls always add operating stress and fatigue on the components. Usually pitch angle is kept at minimum = 0.0 , and any change from this result in drop in torque and hence reduced power production. At sub-rated condition, this is not desirable – so torque is increased by increasing current through rotor control to optimally capture the available power – target is to operate at optimal tip-speed ratio- this can be done through control from the electrical side*

*Many of them got it right*

c)

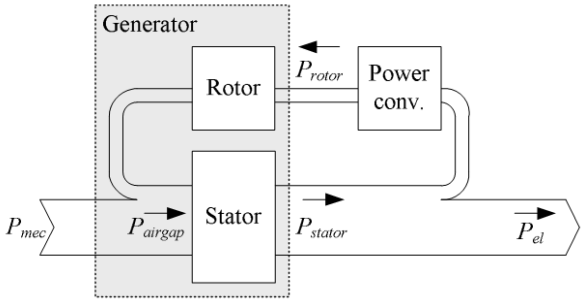
How is it possible to operate a Type-3 (DFIG) machine at lower than synchronous speed (positive slip) in generator mode? [4]

*I expect the student to indicate that direction of rotor current is controllable through converter. At super-synchronous speed neative slip results in current coming out of the rotor. Positive power from rotor slip ring to the grid – in subrated*

*condition – rotor is at subsynchronous speed – so injection of current into rotor of slip frequency will produce net rotor flux which is synchronous with respect to stator flux – two flux systems are locked – so stator produce power to the grid – it is part of that power can be utilised to excite the rotor with slip frequency.. converter control on the rotor side precisely does that at sub-rated condition..*

*[2]*

*They can draw the following diagram also*




[2]

*Almost all got this right*

[25 marks]